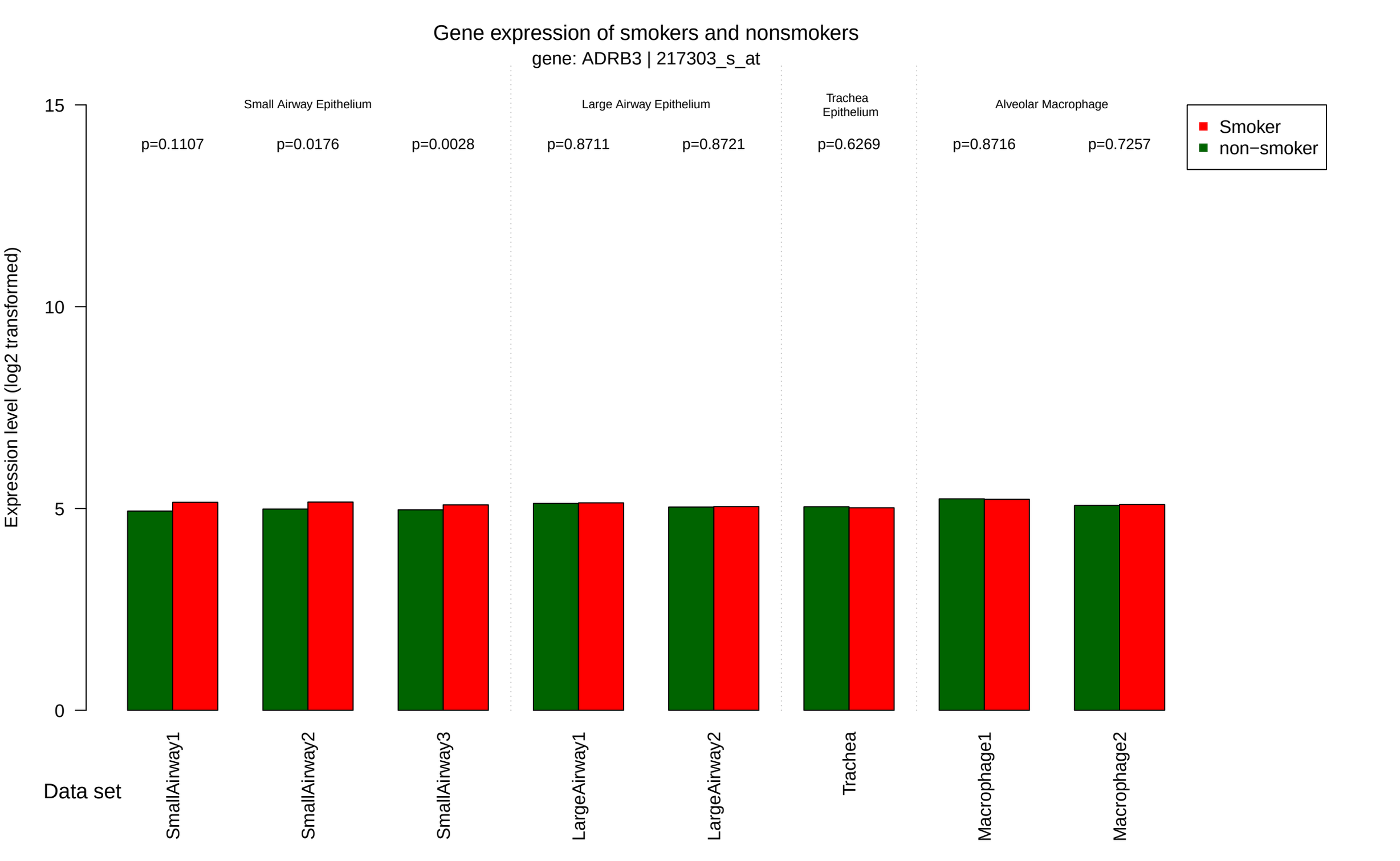
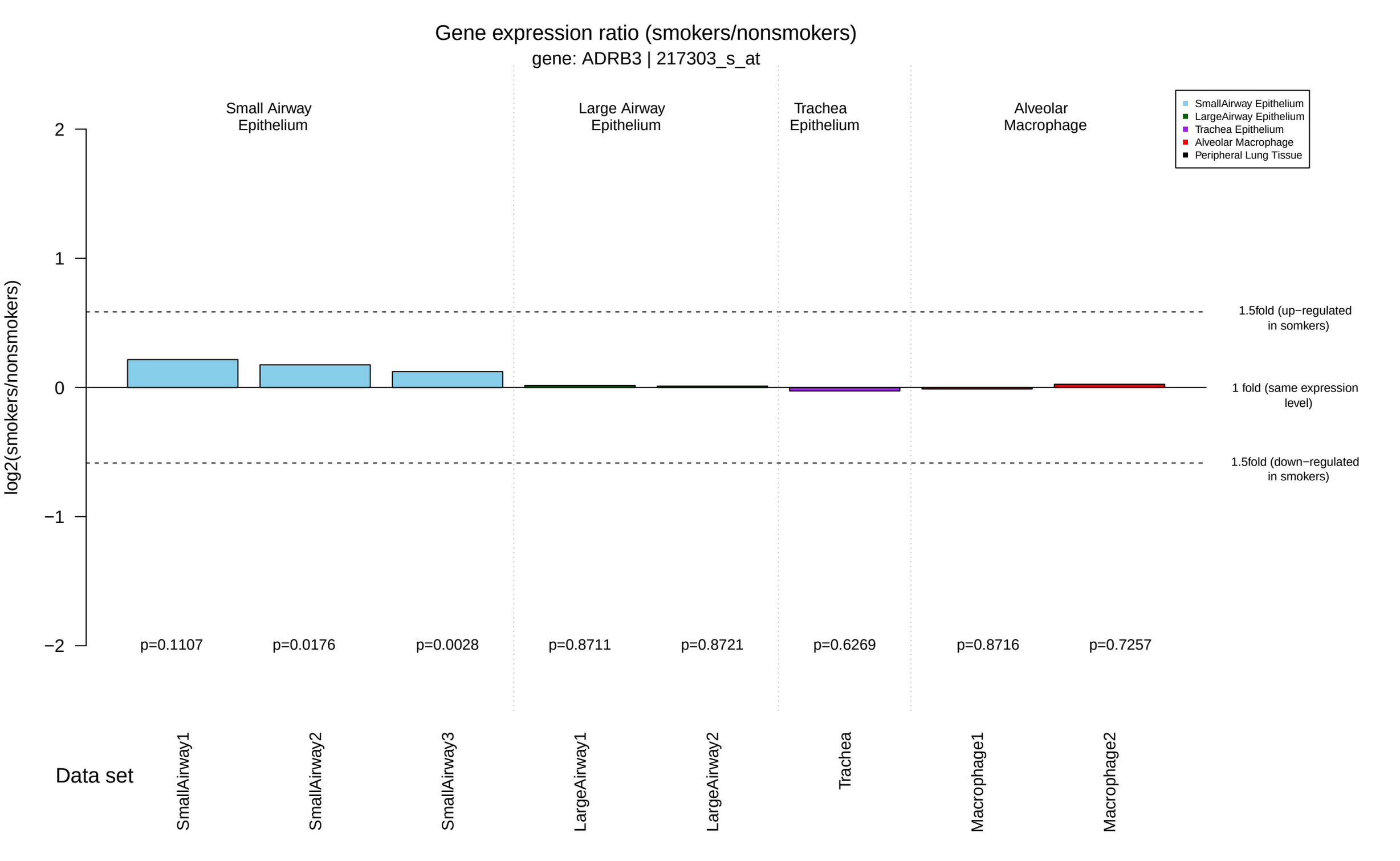
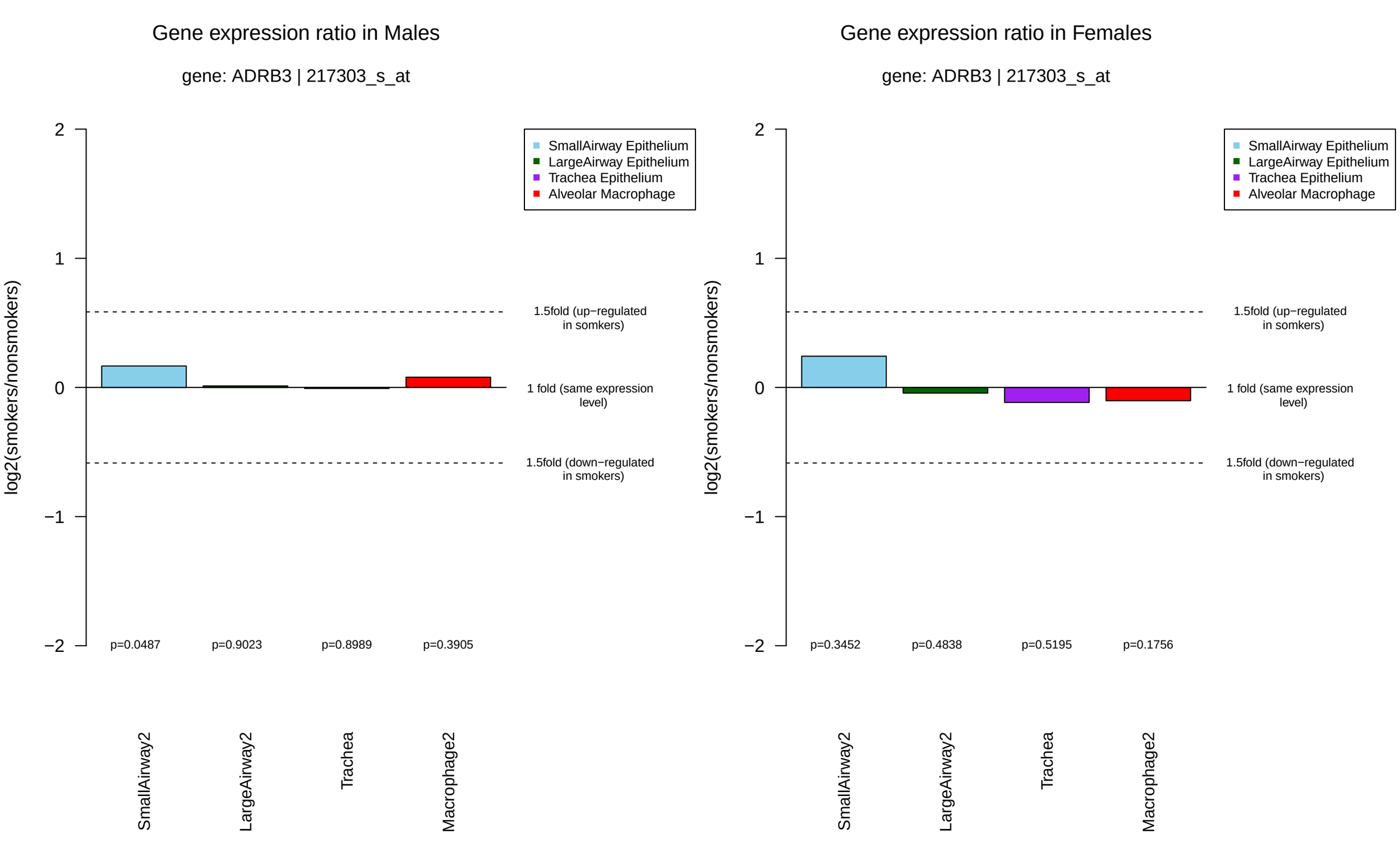
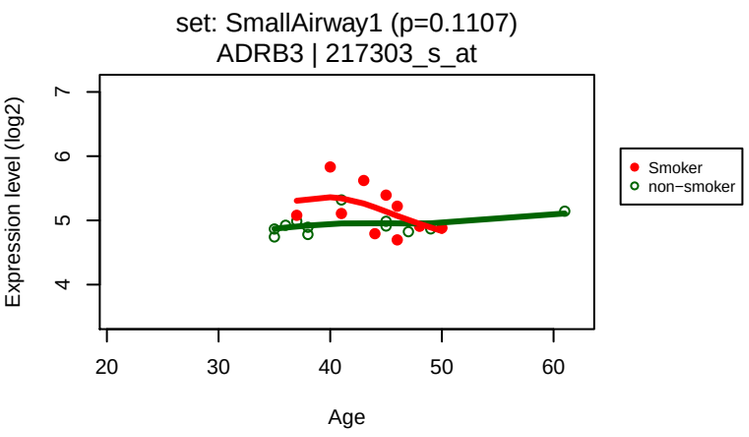
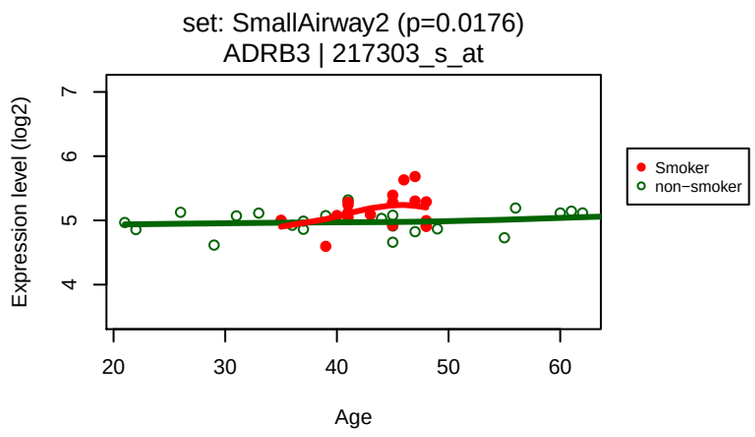
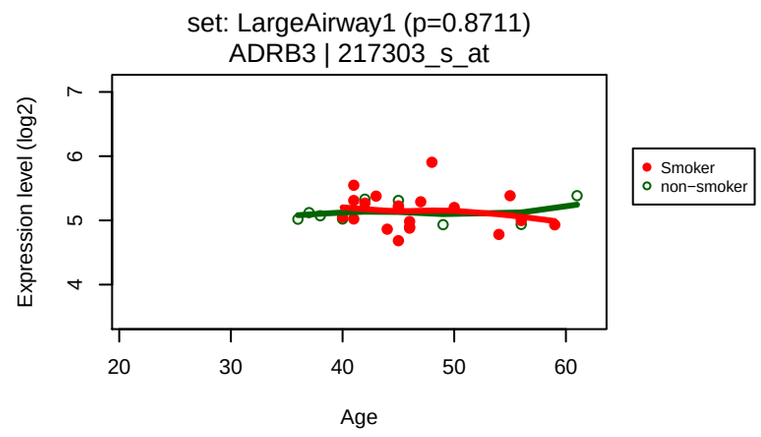

Gene expression of smokers and nonsmokers
gene: ADRB3 | 217303_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1107
p=0.0176
p=0.0028
p=0.8711
p=0.8721
p=0.6269
p=0.8716
p=0.7257
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

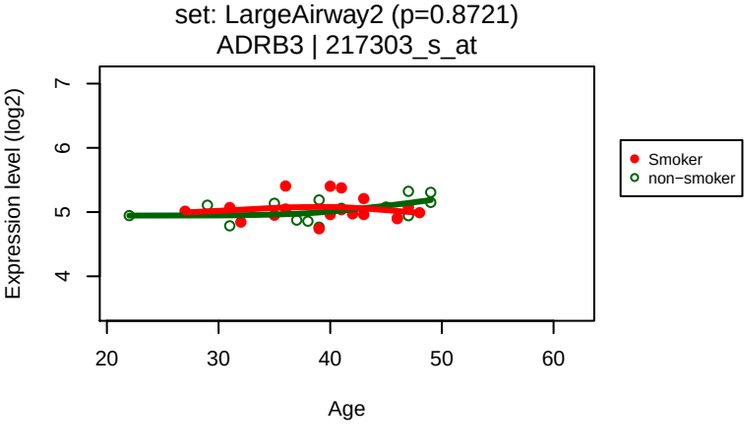

Gene expression ratio (smokers/nonsmokers)
gene: ADRB3 | 217303_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1107
p=0.0176
p=0.0028
p=0.8711
p=0.8721
p=0.6269
p=0.8716
p=0.7257
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

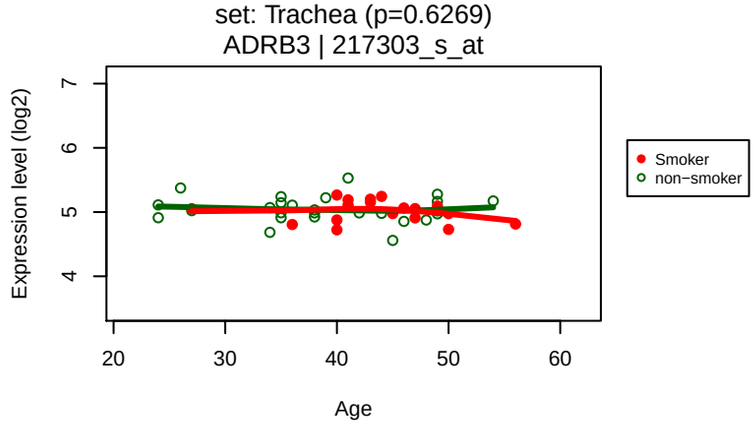

Gene expression ratio in Males
gene: ADRB3 | 217303_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0487
p=0.9023
p=0.8989
p=0.3905
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ADRB3 | 217303_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3452
p=0.4838
p=0.5195
p=0.1756
SmallAirway2
LargeAirway2
Trachea
Macrophage2

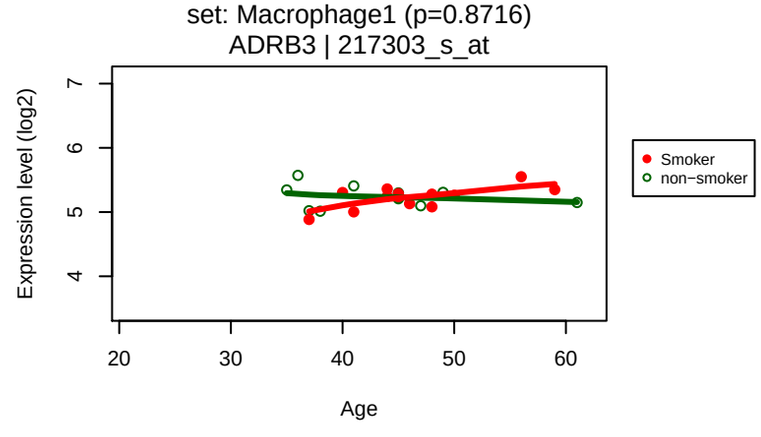

set: SmallAirway1 (p=0.1107)
ADRB3 | 217303_s_at
Expression level (log2)
Age
Smoker
non−smoker

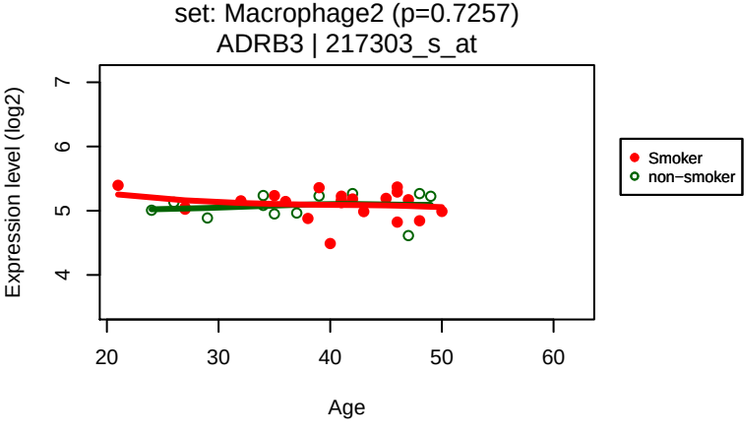

set: SmallAirway2 (p=0.0176)
ADRB3 | 217303_s_at
Expression level (log2)
Age
Smoker
non−smoker

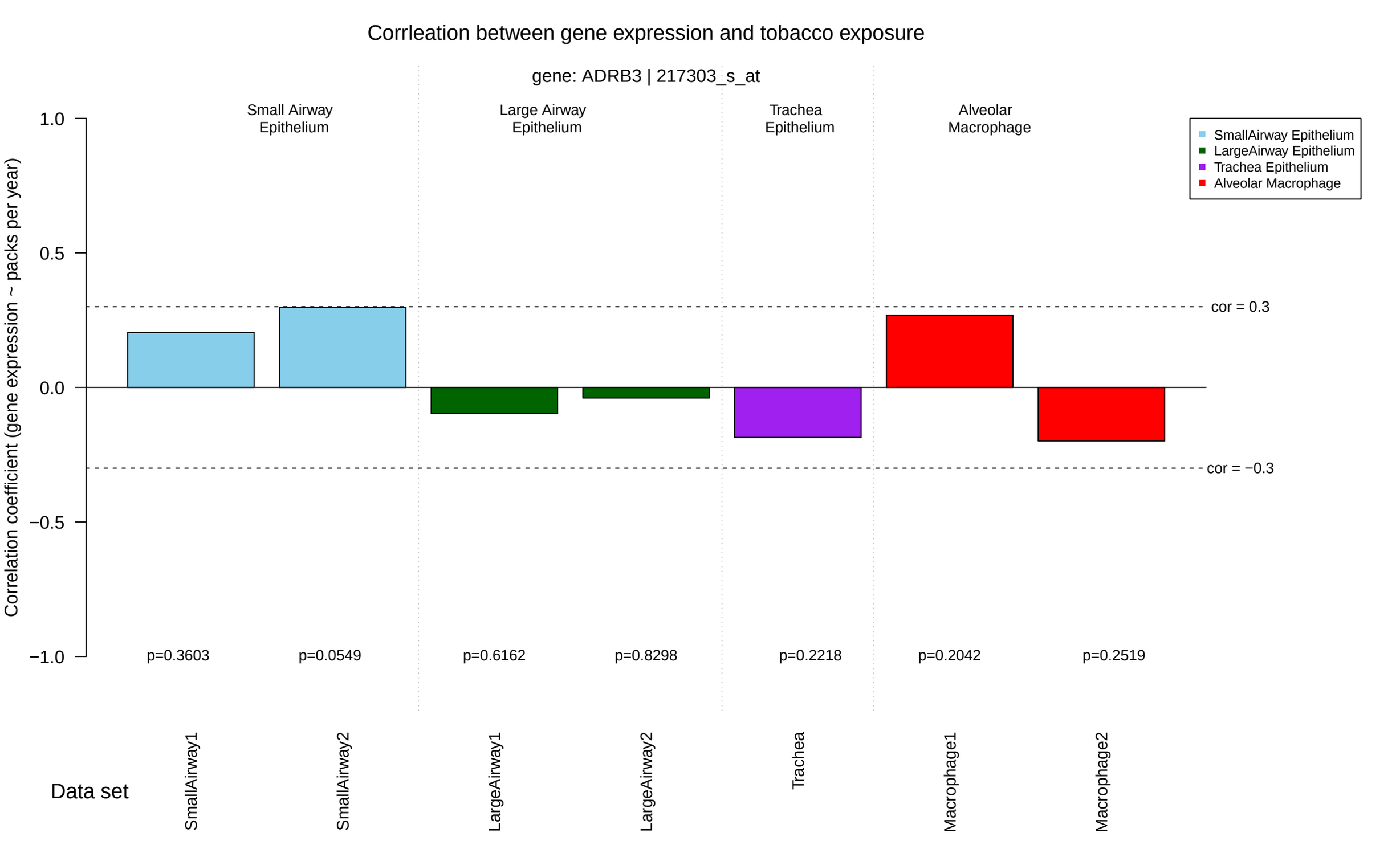

set: LargeAirway1 (p=0.8711)
ADRB3 | 217303_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8721)
ADRB3 | 217303_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6269)
ADRB3 | 217303_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8716)
ADRB3 | 217303_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7257)
ADRB3 | 217303_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ADRB3 | 217303_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3603
p=0.0549
p=0.6162
p=0.8298
p=0.2218
p=0.2042
p=0.2519
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2